

BARUUN-URT, Mongolia

WMO: 443050

Lat: 46.6712N

Lon: 113.2845E

Elev: 3219

StdP: 13.07

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.3	-19.8	-29.5	1.2	-22.1	-25.8	1.5	-18.4	26.0	3.4	21.8	4.4	6.0	320	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.8	65.0	86.8	63.6	83.3	62.3	69.0	84.1	66.9	80.9	65.1	77.5	10.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	102.1	73.7	62.6	95.6	71.8	60.7	89.5	71.0	35.3	83.9	33.5	80.9	31.9	77.7	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	23.3	20.6	-27.3	96.7	3.8	4.4	-30.1	99.9	-32.3	102.5	-34.5	105.0	-37.3	108.2
			-27.3	71.3	4.2	3.9	-30.4	74.1	-32.8	76.4	-35.2	78.5	-38.3	81.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	-5.7	3.3	21.6	40.4	54.0	65.5	70.7	67.2	55.2	37.6	15.5	-0.4	(6)
(7)		DBStd	28.13	8.54	10.14	11.47	9.39	8.87	7.39	5.88	6.49	8.53	9.21	11.74	8.41	(7)
(8)		HDD50	7321	1727	1307	879	311	60	1	0	0	44	394	1036	1562	(8)
(9)		HDD65	11170	2192	1727	1344	738	355	81	11	51	309	849	1486	2027	(9)
(10)		CDD50	2059	0	0	0	23	184	467	642	532	200	11	0	0	(10)
(11)		CDD65	432	0	0	0	0	14	97	188	118	15	0	0	0	(11)
(12)		CDH74	4446	0	0	0	11	273	1094	1738	1132	197	1	0	0	(12)
(13)		CDH80	1575	0	0	0	1	78	400	669	383	44	0	0	0	(13)
(14)	Wind	WSAvg	8.5	7.2	7.3	9.1	11.1	11.3	9.1	8.0	7.7	8.3	8.4	7.5	7.1	(14)
(15)	Precipitation	PrecAvg	7.6	0.1	0.1	0.1	0.2	0.6	1.5	2.3	1.7	0.8	0.2	0.1	0.1	(15)
(16)		PrecMax	14.7	0.6	0.2	0.6	0.9	2.8	4.1	6.5	4.2	3.4	0.8	0.4	1.9	(16)
(17)		PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.1	0.1	0.1	0.2	0.6	1.0	1.4	1.0	0.7	0.2	0.1	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	35.3	58.2	74.4	87.9	93.5	96.9	94.3	85.1	69.7	47.8	25.7	(19)
MCWB			17.2	29.3	40.4	50.3	57.1	64.1	69.1	67.2	61.2	51.2	36.6	22.3	(20)	
2%		DB	14.5	28.2	50.3	67.7	81.0	88.8	91.4	88.2	79.6	62.3	41.5	19.1	19.1	(21)
		MCWB	12.3	24.2	37.1	48.0	56.1	63.2	67.0	64.4	58.4	47.1	33.4	16.8	16.8	(22)
5%		DB	10.4	22.7	44.2	62.2	76.0	84.6	87.1	84.0	74.7	56.7	36.3	15.1	15.1	(23)
		MCWB	8.9	19.9	33.9	44.6	53.6	61.7	65.7	62.4	56.4	43.8	29.9	13.3	13.3	(24)
10%		DB	6.6	18.5	38.7	57.3	71.5	80.3	83.1	80.3	70.2	52.3	31.5	11.7	11.7	(25)
		MCWB	5.4	16.3	30.5	42.4	51.6	60.0	64.3	61.4	54.3	41.0	26.7	10.3	10.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	29.7	42.1	53.0	62.3	69.2	73.3	70.4	65.5	52.9	37.8	22.4	(27)
MADB			19.9	34.3	56.4	69.4	80.6	87.7	90.9	87.0	77.3	67.4	46.2	25.6	25.6	(28)
2%		WB	12.5	24.1	37.5	49.0	57.9	66.1	70.0	67.5	60.6	47.5	33.5	16.8	16.8	(29)
		MADB	14.4	28.1	49.3	65.1	75.8	82.7	84.6	83.1	74.0	59.5	41.2	18.9	18.9	(30)
5%		WB	8.9	19.9	34.1	45.7	55.2	63.7	68.0	65.5	58.2	44.8	30.0	13.3	13.3	(31)
		MADB	10.3	22.7	43.6	61.1	72.2	78.6	81.4	78.4	71.4	55.7	35.9	14.9	14.9	(32)
10%		WB	5.4	16.4	30.9	42.6	52.7	61.5	66.2	63.7	55.7	41.6	26.8	10.3	10.3	(33)
		MADB	6.5	18.6	38.3	56.9	69.1	75.6	78.7	75.1	67.7	51.4	31.1	11.6	11.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	18.6	20.6	22.8	24.0	20.9	19.5	20.8	22.1	20.6	17.6	14.8	(35)
MCDBR			18.0	21.3	27.0	30.7	31.1	27.2	25.4	26.3	27.8	26.4	21.9	17.5	17.5	(36)
MCWBR			16.4	18.0	17.3	17.8	15.7	11.5	10.2	10.9	13.9	16.1	15.9	15.7	15.7	(37)
5% WB		MCDBR	17.9	21.2	25.8	28.9	27.3	23.8	21.8	22.2	24.4	24.8	21.7	17.5	17.5	(38)
		MCWBR	16.4	18.2	16.9	18.2	15.9	12.0	10.8	11.2	13.5	16.2	16.0	15.7	15.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.253	0.275	0.315	0.373	0.436	0.453	0.411	0.379	0.340	0.305	0.270	0.256	(40)	
(41)		taud	2.328	2.287	2.249	2.135	1.992	2.056	2.251	2.310	2.354	2.380	2.359	2.302	(41)	
(42)		Ebn at Noon	271	285	288	279	263	259	269	274	276	270	260	253	(42)	
(43)		Edh at Noon	23	30	36	45	54	51	42	38	33	27	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	653	1015	1476	1863	2102	2090	1996	1844	1506	1068	685	514	(44)	
(45)		RadStd	40	55	47	70	112	141	90	98	86	59	41	36	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) -3.67	(g) N/A	(h) -2.27	(i) -2.27	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	-2.54	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page